

Astrophotography: First Light

Three days, one weekend, no telescope. Arrive with a phone or a DSLR and a tripod; leave with a stacked Milky Way nightscape you processed yourself.

7	16	17 h	6 h	Astrophotography Group
SESSIONS	CONCEPTS TAUGHT	TOTAL	OUTDOORS	AUDIENCE

Moon: start when the moon is **0–2.5 days old** (🌑 new → 🌒 waxing crescent) or start when the moon is **26–29.5 days old** (🌓 waning crescent → 🌑 new)

ASSUMED ALREADY KNOWN

Lenses, Mirrors & Focal Length

Lunar Observing

The Milky Way Band

Light Pollution

The Pole Star

Star Charts & Planetarium Apps

The Eye as a Detector

Closed under prerequisites: 28 concepts in total are taken as prior knowledge.

1 Day 1 — The camera, before dark

CLASSROOM · 2 H 30 MIN · DAY 1

KIT Projector · DSLR / mirrorless · Smartphone · Tripod

L3 Exposure: Aperture, Shutter, ISO

Three controls set how much light a camera records and how noisy it looks. At night you run all three near their limits — astrophotography is exposure at the extreme.

L3 RAW Files & Bit Depth

JPEG throws away the faint data astrophotography lives on. RAW keeps every bit the sensor recorded — mandatory before any stacking or stretching.

L3 Reading the Histogram

The histogram is the only honest exposure meter at night — the LCD preview lies. Expose so the sky peak sits clear of the left wall without clipping stars.

TRAINER NOTE In daylight, in a room, on their own cameras — everyone finds manual mode, RAW and the histogram before the sun goes down. A participant hunting for manual mode at 9pm in the dark has lost the night, and this session exists to prevent exactly that. Make them shoot one deliberately blown frame and one crushed one, and read both off the histogram.

2 Day 1 — Why night shots fail

CLASSROOM · 1 H 30 MIN · DAY 1

KIT Projector · Laptop

L2 Dark Adaptation & Averted Vision

Eyes take 20–30 minutes to reach full night sensitivity, and one white-light glance resets them. Looking slightly beside a faint object uses your more sensitive off-centre retina.

L3 Focusing in the Dark

Autofocus is useless on stars. Live-view at 10× on a bright star, or a Bahtinov mask's diffraction spikes, gets focus exact — the single most common beginner failure.

L4 How Sensors Work

Pixels convert photons to electrons; quantum efficiency, pixel size, full-well depth and read noise decide astro performance. Why cooled mono cameras exist.

L4 Signal, Noise & SNR

Shot noise, read noise, thermal noise versus faint signal: every astro-imaging decision — exposure length, ISO, cooling, stacking — is an SNR calculation in disguise.

TRAINER NOTE Noise is not a defect of cheap cameras, it is photon statistics — which is why the fix is more total exposure, not a better body. That framing makes stacking on Day 2 an obvious consequence rather than a software trick. Practise focusing on a distant streetlight now; live-view on a star at midnight is a different sport.

3 Day 1 — First night: tripod only

IMAGING NIGHT · 3 H · DAY 1

SKY needs **Dark sky (no moon)** · Milky Way, Star trails, Constellations

KIT DSLR / mirrorless · Smartphone · Tripod · Red torch · Power bank / batteries

L2 Smartphone Astrophotography

A phone held (or bracketed) at the eyepiece captures the Moon and planets; night modes catch constellations and even the Milky Way. The zero-cost gateway to imaging.

L3 Nightscape Photography

Wide-angle sky-over-landscape shots from a fixed tripod. The NPF/500 rule caps exposure before stars trail — beyond it, you need tracking.

L3 Star Trail Photography

Stack hundreds of fixed-tripod frames and stars draw arcs around the celestial pole — diurnal motion made visible and beautiful in one image.

TRAINER NOTE Teach the 500 rule, then make them break it and pixel-peep the trailing — the number means nothing until they have watched it fail. Modern phone night mode genuinely works; do not be snobbish about it, it is the camera most of them will keep using. Everyone shoots trails as a stack of subs rather than one long exposure, because Day 2 needs those frames.

4 Day 2 — Stacking: the whole trick

INDOOR LAB · 3 H · DAY 2

KIT Laptop · Power bank / batteries

L4 Darks, Flats & Bias

Dark frames map thermal signal, flats map dust and vignetting, bias maps readout offset. Calibration subtracts the camera's own fingerprints from your data.

L4 Image Stacking

Averaging N aligned frames cuts noise by $\sqrt{N}$ while signal stays put — the single most powerful trick in astrophotography. Software: DeepSkyStacker, Siril, AutoStakkert.

TRAINER NOTE Their own frames from last night, never a demo set — the payoff is watching their own noisy sub become a clean image. Darks and flats are the two that matter; bias can wait a lifetime. Someone will have forgotten flats: let the gradient show, then shoot flats against a laptop screen and redo it. That failure teaches flats better than a slide.

5 Day 2 — Second night: the sky moves

IMAGING NIGHT · 3 H · DAY 2

SKY needs **Dark sky (no moon)** · Milky Way, Star trails, Constellations

KIT DSLR / mirrorless · Smartphone · Tripod · Red torch · Power bank / batteries · Star chart / planisphere

L2 Observing Logs & Sketching

Date, time, sky conditions, instrument, and what you actually saw — a log turns stray sessions into a record of growth, and sketching trains the eye to see more.

L3 Sky Timelapse

Hundreds of frames, seconds apart, compressed into video: the sky wheels, the Milky Way rises, satellites streak. Diurnal motion becomes something you can show, not tell.

TRAINER NOTE Night one was about one frame; tonight is about a sequence. An intervalometer run doubles as timelapse frames and as stacking subs, so nobody chooses. This is also the night to log properly — settings, target, conditions — because the next clear sky should start from knowledge, not from scratch. If a tracker is on site, demonstrate it; do not teach it, it is the next course.

6 Day 3 — Processing, and stopping in time

INDOOR LAB · 3 H · DAY 3

KIT Laptop · Projector

L5 Post-Processing

Stretching, gradient removal, colour calibration, noise control — developing the stacked data into a finished image. Half the craft happens after the telescope is packed.

TRAINER NOTE Stretching is the one idea: the data is linear, the eye is not. Gradient removal, then colour balance. The hardest thing to teach is when to stop — put an over-processed image beside a restrained one without saying which is which, and let the room call it.

Citizen Science

Variable-star estimates (AAVSO), occultation timings, meteor counts, comet hunting: organised programmes where amateur measurements feed real research.

TRAINER NOTE Everyone puts one image up with its settings attached. Close on citizen science so the weekend has an afterwards: variable-star estimates and meteor counts need exactly the kit they now own and know how to use.

FULL KIT LIST

DSLR / mirrorless	sessions 1, 3, 5
Laptop	sessions 2, 4, 6
Smartphone	sessions 1, 3, 5
Power bank / batteries	sessions 3, 4, 5
Projector	sessions 1, 2, 6, 7
Red torch	sessions 3, 5
Star chart / planisphere	sessions 5
Tripod	sessions 1, 3, 5